

3. Metode Maximum Likelihood

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1. Definisikan fungsi likelihood $L(\theta)$
2. Mungkin lebih mudah dgn $\log \ell(\theta)$
3. Jika diperoleh, $L'(\theta) = 0$ *~ sudah dapat penduga*
4. Pastikan pd titik $\hat{\theta}$, $L(\theta)$ maksimum $\rightarrow L''(\theta) < 0$

ex/ $X_1, X_2, \dots, X_n \sim \text{iid}$ Eksponensial (λ)

$$pdf: f(x|\lambda) = \lambda e^{-\lambda x}, \quad x \geq 0, \lambda > 0$$

Tentukan MLE untuk λ !

① Tentukan fungsi likelihood

$$L(\lambda) = \prod_{i=1}^n f(x_i|\lambda)$$

$$= \prod_{i=1}^n \lambda e^{-\lambda x_i}$$

$$= \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

② $\log L(\lambda)$

$$\Rightarrow \ell(\lambda) = \log(\lambda^n e^{-\lambda \sum x_i})$$

$$= n \log \lambda - \lambda \sum_{i=1}^n x_i$$

③ $L'(\lambda) = 0$

$$\Rightarrow \frac{d}{d\lambda} \ell(\lambda) = \frac{n}{\lambda} - \sum x_i = 0$$

$$\frac{n}{\lambda} = \sum x_i$$

$$\lambda = \frac{n}{\sum x_i}$$

$$= \frac{1}{\bar{x}}$$

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\sum x_i = n \cdot \bar{x}$$

④ $L''(\lambda) < 0$

$$\Rightarrow \frac{d}{d\lambda} \left(\frac{n}{\lambda} - \sum x_i \right) = -\frac{n}{\lambda^2} < 0 \rightarrow \text{Sehingga } \hat{\lambda}_{MLE} = \frac{1}{\bar{x}}$$

↳ benar maksimum

ex/ $X_1, X_2, \dots, X_n \sim \text{iid}$ Poisson (λ)

Tentukan MLE untuk λ !

$$pdf: f(x|\lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\textcircled{1} L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$= \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod_{i=1}^n x_i!}$$

② $\log$ likelihood

$$\log L(\lambda) = \ell(\lambda) = \log \left(\frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod_{i=1}^n x_i!} \right)$$

$$= -n\lambda + \sum x_i \log(\lambda) - \sum_{i=1}^n \log(x_i!)$$

③ $\ell'(\lambda) = -n + \sum x_i \cdot \lambda^{-1} = 0$

$$n = \frac{\sum x_i}{\lambda}$$

$$\lambda = \frac{\sum x_i}{n}$$

$$\hat{\lambda}_{MLE} = \bar{x}$$

④ cek maksimum $\Rightarrow \ell''(\lambda) < 0$

$$\hookrightarrow \frac{d}{d\lambda} (-n + \sum x_i \cdot \lambda^{-1}) = -\frac{\sum x_i}{\lambda^2} < 0$$

→ sehingga $\hat{\lambda}_{MLE} = \bar{x}$

ex/ $X_1, X_2, \dots, X_n \sim \text{Uniform}(0, \theta)$

$$pdf: f(x|\theta) = \frac{1}{\theta}, \quad 0 \leq x \leq \theta$$

Tentukan MLE untuk θ

① Fungsi likelihood

$$L(\theta) = \prod_{i=1}^n f(x_i|\theta) = \prod_{i=1}^n \frac{1}{\theta} \cdot 1_{\{0 \leq x_i \leq \theta\}}$$

$$= \frac{1}{\theta^n} \cdot 1_{\{\theta \geq \max x_i\}}$$

$$= \frac{1}{\theta^n} \cdot 1_{\{\theta \geq \max x_i\}}$$

② $\log L(\theta) = \log \left(\frac{1}{\theta^n} \cdot 1_{\{\theta \geq \max x_i\}} \right)$

$$= -n \log(\theta)$$

③ $\frac{d}{d\theta} \ell(\theta) = -\frac{n}{\theta}$ *∵ karena $\ell'(\theta) < 0$*

maka log likelihood selalu menurun thdp θ

④ $\ell''(\theta) = \frac{n}{\theta^2} > 0$ *→ bentuk cekung ke atas. jadi, tdk ada max di interior domain. sehingga nilai max dicapai ketika batas terkecil domain valid,*

$$\text{yaitu } \hat{\theta}_{MLE} = \max\{x_1, x_2, \dots, x_n\}$$

$$\text{atau } \hat{\theta}_{MLE} = \max(x_i) = x_n.$$

MLE Eksponensial pakai Newton-Raphson

Model: $X_1, \dots, X_n \sim \text{iid Exp}(\lambda)$, $f(x|\lambda) = \lambda e^{-\lambda x}$, $x \geq 0$

Data contoh: $x = \{2.1, 0.7, 1.5, 0.9, 1.8\}$

Ringkasan: $n=5$, $\sum x_i = 6.9$, $\bar{x} = \frac{6.9}{5} = 1.38$

Tujuan: cari $\hat{\lambda}$ yg memaksimumkan likelihood